

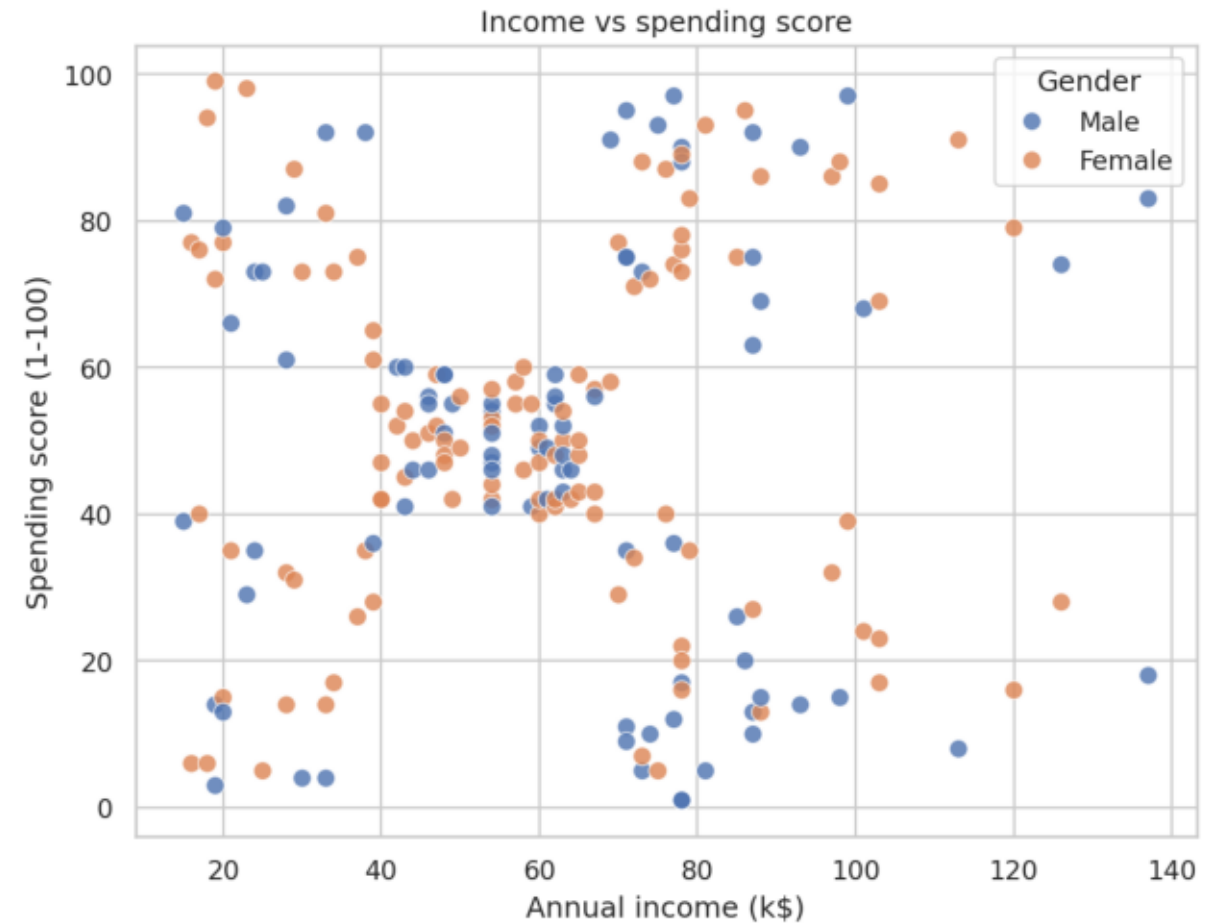
1. The question

- Client: a shopping mall's marketing team.
- Today every member gets the same promotion.
- Need: split members into a few groups so each gets a campaign that fits.
- Unsupervised task: no target; success = well-separated, actionable groups (silhouette score).

- Kaggle Mall Customer Segmentation: 200 customers × 5 columns.
- Age, gender, annual income (k\$), spending score (1-100).
- Biggest data problem fixed: misspelled 'Genre' and column names with spaces/brackets → renamed.
- No missing values or duplicates; 2 high-income outliers kept (real customers).

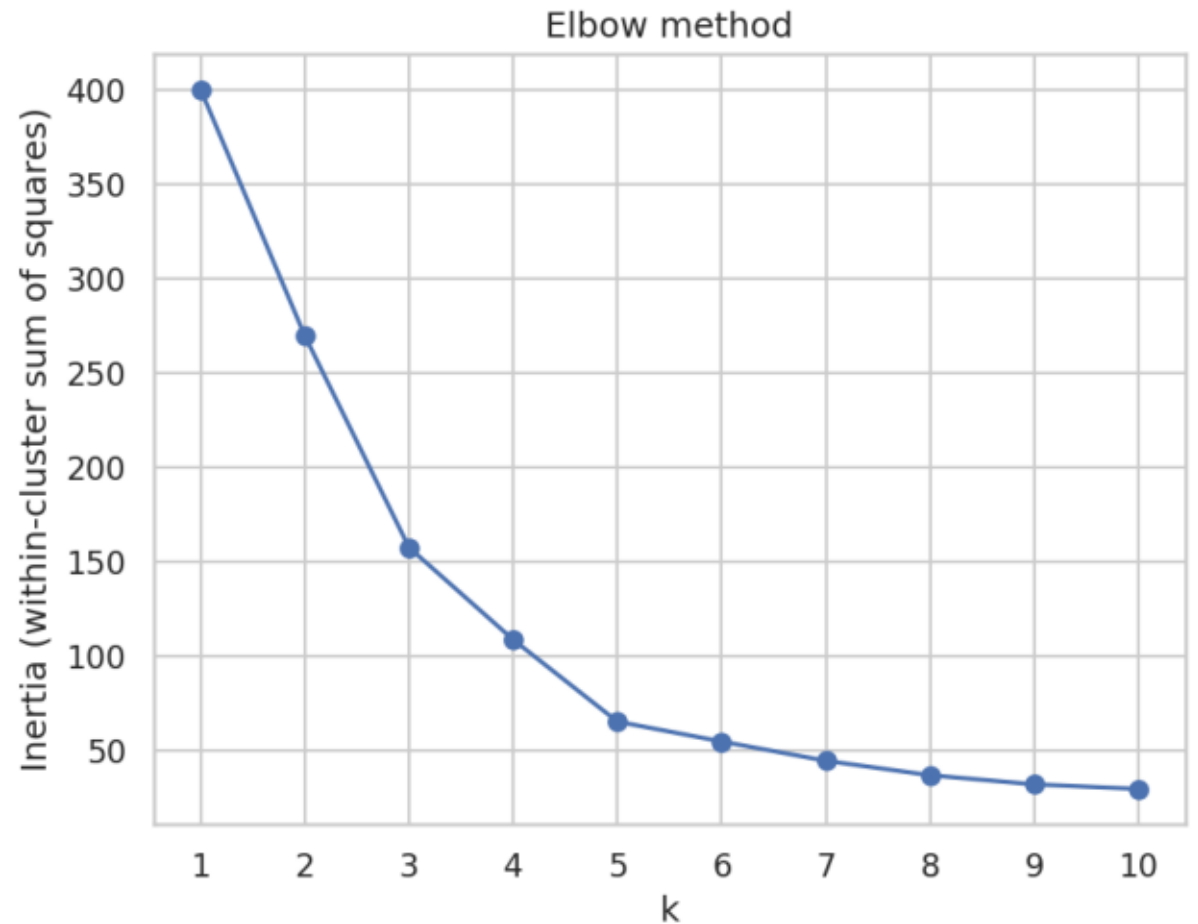
3. Key insights

- Income does not predict spending (correlation 0.01) → we must segment.
- By eye: one middle group + four corner groups.
- All 62 high spenders are 40 or younger.



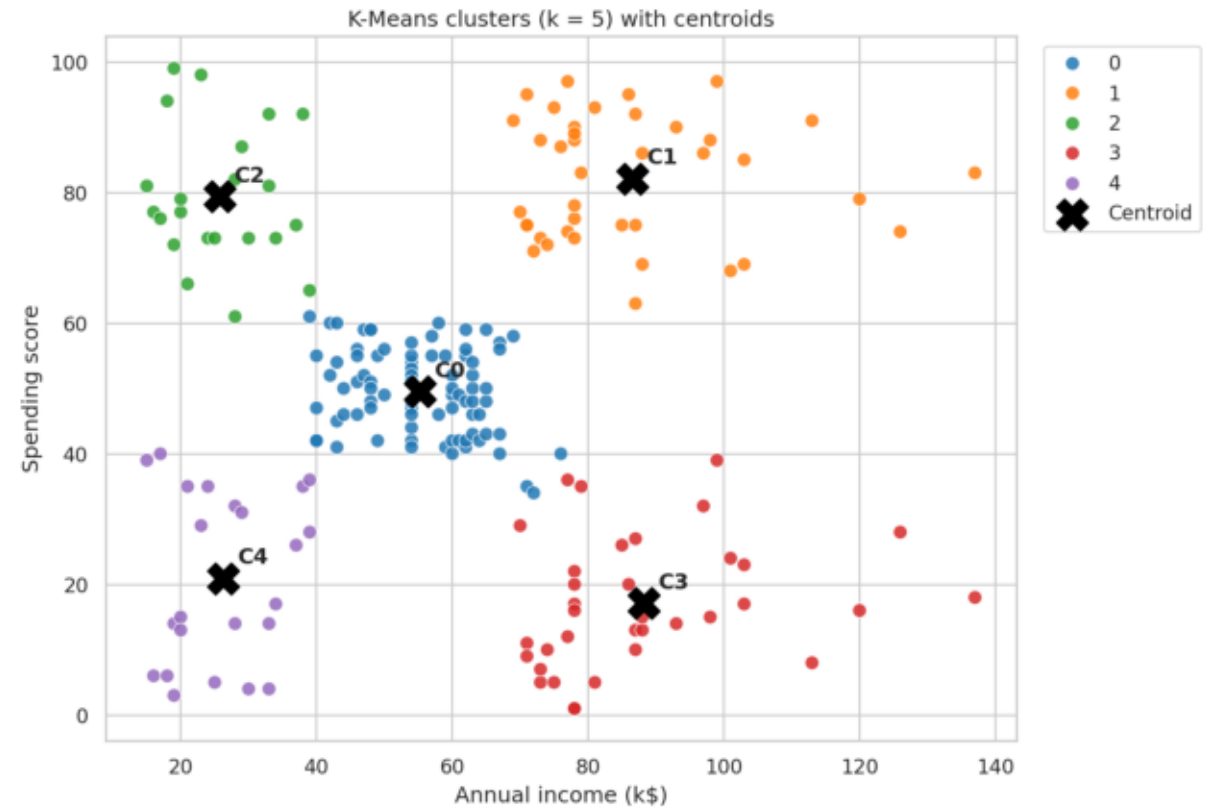
4. The model

- Baseline: split at median income and median spending → 4 groups, silhouette 0.332.
- Final: K-Means on scaled income + spending, random_state=42.
- $k = 5$: elbow bends at 5 and silhouette is highest at 5.
- Why K-Means: simple, fast, centroids = 'typical customer' easy to explain.



5. Results

- K-Means silhouette 0.555 vs 0.332 baseline.
- Five personas: Premium Spenders (39), Careful High Earners (35), Average Regulars (81), Young Trend Seekers (22), Budget-Conscious (23).
- Same groups with any random seed; only 9 borderline customers.



- Replace one promotion with five targeted campaigns.
- Biggest budget: Careful High Earners (high income, low spend = growth).
- Retain Premium Spenders with a VIP tier.
- Limits: 200 customers, unknown spending-score formula; A/B-test before roll-out.